

Advanced Statistical Methods

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# Non-Parametric Statistics

Complete Lecture Notes

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Weeks 01 - 07

[0.3cm] CDF Inference · Multiple Samples

Sign Test · Wilcoxon Signed Rank · Wilcoxon Rank Sum  
Mann-Whitney · Kruskal-Wallis · Kolmogorov-Smirnov

Distribution-Free Methods for Statistical Inference

# **Non-Parametric Statistics: Complete Lecture Notes**

Advanced Statistical Methods (ASM-I)

Weeks 01-07

*These notes compile lectures on non-parametric and distribution-free methods for hypothesis testing and estimation.*

Compiled from lecture slides, handouts, and class notes.

All mathematical content is presented for educational purposes.

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# Chapter 1

## Introduction to Non-Parametric Methods

### 1.1 Recap: Parametric Methods

Before introducing non-parametric methods, we briefly review the standard parametric tests that motivate the need for distribution-free alternatives.

#### 1.1.1 One-Sample Mean

- **Assumed Model:**  $\mathbb{E}[X_i] = \mu$
- **Hypothesis Tests:**

$$H_0 : \mu = \mu_0 \quad H_a : \mu \neq \mu_0$$

- **Test Statistic:**

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

- **Assumptions:**

- $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F$
- If  $n$  is large,  $F$  can be any distribution (CLT applies).
- If  $n$  is small, we assume  $F \sim \mathcal{N}$  (Normal Distribution).

- **Open question:** What if  $n$  is small and  $F$  is not Normal?

#### 1.1.2 Two-Sample Mean

- **Samples:**  $X_1, \dots, X_n \sim F$  and  $Y_1, \dots, Y_m \sim G$
- **Assumption:**  $\mathbb{E}[X] = \mu_1, \mathbb{E}[Y] = \mu_2$
- **Hypothesis:**

$$H_0 : \mu_1 = \mu_2 \quad H_a : \mu_1 \neq \mu_2$$

- **Conditions:**

- If  $F$  and  $G$  are both Normal  $\Rightarrow$  2-sample  $t$ -test.
- If both  $m$  and  $n$  are large  $\Rightarrow$  Normal test.

### 1.1.3 Variance Test (One Sample)

• **Sample:**  $X_1, \dots, X_n \sim F$

• **Variance:**  $\text{Var}(X_i) = \sigma^2$

• **Statistic:**

$$\frac{(n-1)S^2}{\sigma_0^2} \sim \chi_{n-1}^2$$

•  $S^2$  is the unbiased sample variance.

• **Hypothesis:**  $H_a : \sigma^2 \neq \sigma_0^2$

• **Assumption:**  $F$  is Normal.

### 1.1.4 Two-Sample Variance

• **Statistic:**  $S_1^2/S_2^2$

• **Hypothesis:**  $H_0 : \sigma_1^2 = \sigma_2^2$

• **Test:**  $F$ -test

• **Assumption:**  $F = \mathcal{N} = G$  (both distributions are Normal).

### 1.1.5 ANOVA (One-Way)

• Tests equality of means of several samples.

• Valid under Normal assumptions.

## 1.2 Motivation for Non-Parametric Tests

#### Remark

**Central question:** What if the sample size  $n$  is small and the parent distribution is unknown?

• If the parent distribution is *known* but not Normal, we may be able to construct the distribution of the test statistic using probability theory. For example:

$$- F \sim \text{Exp}(\lambda) \Rightarrow \sum X_i \sim \text{Gamma}(n, \lambda)$$

$$- F \sim \text{Pois}(\lambda) \Rightarrow \sum X_i \sim \text{Pois}(n\lambda)$$

Then we can construct the test statistic under  $H_0$  using knowledge of probability theory.

• However, *often we do not know the distribution.*

**Definition: Distribution-Free Tests**

The tests we shall construct are **distribution-free**. We do not assume any parametric model a model with known form but a finite number of unknown parameters.

**1.2.1 Why Not Always Use Non-Parametric Methods?**

The flexibility of non-parametric methods is attractive but comes with a cost:

- When a parametric model is true/correct, using that parametric test is “better”.
- **Power:** For hypothesis testing, both parametric and non-parametric tests will have the same significance level  $\alpha$ . But the parametric test (under the correct parent model) will have more power.
- **Confidence Intervals:** Both procedures will give the same average probability (coverage), but the parametric interval will be shorter/narrower (more precise).

**1.3 The Sign Test (One-Sample Median Test)****Remark**

Use non-parametric techniques if the distribution of the data is unknown. The Sign Test is the non-parametric alternative to the one-sample  $t$ -test.

**1.3.1 Concept**

- Generally for  $F$ , the mean may not exist (e.g., Cauchy distribution), but the **Median always exists**.
- Median equals Mean if  $F \approx \mathcal{N}$  (Normal).

**Definition: Median**

The **median**  $M$  of a distribution  $F$  is defined as:

$$M = \min\{a : F(a) \geq \frac{1}{2}\}.$$

### 1.3.2 Example Scenario: Arsenic Concentration

#### Example: Arsenic in Drinking Water

**Source:** StatLib (CMU) Arsenic in drinking water.

**Data:**  $X_1, \dots, X_{21}$  (Arsenic concentration in toenail clippings of 21 individuals).

**Hypothesis:**

$$H_0 : \theta = 0.2 \quad \text{vs} \quad H_a : \theta > 0.2 \quad (\text{Right-tailed})$$

where  $\text{med}(X_i) = \theta$ .

### 1.3.3 Procedure

#### Procedure: Sign Test Procedure

1. Under  $H_0$ , observe that  $P(X_i > 0.2) = \frac{1}{2}$ .
2. Sample  $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F$  (independent).
3. Define the indicator variable  $Y_i$ :

$$Y_i = \begin{cases} 1 & \text{if } X_i > \theta_0 \\ 0 & \text{if } X_i \leq \theta_0 \end{cases}$$

4. Test Statistic:

$$Y = \sum_{i=1}^n Y_i \sim \text{Bin}(n, \frac{1}{2})$$

(Since  $X_i$  are independent,  $Y_i$  are independent Bernoulli trials.)

### 1.3.4 Rejection Region

- Reject for *large* values of  $Y$ .
- $P(Y > C) = \alpha$  (Right-tailed).
- Depending on the type of  $H_a$ , a left-tailed or two-tailed rejection region may be needed.

### 1.3.5 Sign Test Analysis

#### Remark

A **randomness check** is necessary before applying the sign test.

**Example: Sign Test Calculation**

$H_a : \theta > 0.2$ , observed  $y = 10$  with  $n = 21$ .

**P-value calculation (in R):**

```
2.pbinom(10, 21, 0.5)    # gives 1
```

If p-value  $> \alpha$ , do *not* reject  $H_0$ .

**1.3.6 Large-Sample Approximation for the Sign Test**

- If  $n$  is large, use the Normal approximation to the Binomial (converges to normal quickly;  $n \geq 10$  is usually sufficient). Note: convergence to a general  $F$  may require a much larger  $n$ .
- Test Statistic:  $Y \sim \text{Bin}(n, 0.5)$ .
- With  $n = 21$ ,  $y = 10$ :

$$Z = \frac{Y - np}{\sqrt{npq}} = \frac{Y - 0.5n}{0.5\sqrt{n}}$$

- In R:

```
2 * pnorm(Z)    # gives 0.8
```

- Do not reject  $H_0$  at any  $\alpha < 0.8$ .

**1.3.7 Summary of the Sign Test****Theorem: Sign Test Summary**

- A one-sample median test looking at the *sign* of  $X_i - \theta_0$ .
- Widely applicable. The only assumption is that  $F$  is **continuous**.
- **Drawback:** Low power under many alternatives. It ignores the *magnitude* of the difference.
- Example: A density could be symmetric but shaped such that signs are balanced even if the bulk of data is far from  $\theta_0$ .

**1.4 Wilcoxon Signed Rank Test****Definition: Wilcoxon Signed Rank Test**

- Assumes  $F$  is **continuous and symmetric**.
- Ranks are distribution-free.
- **Idea:** Give more weight to  $X_i$  that are much larger than  $\theta_0$  (large deviations receive larger weight).

### 1.4.1 Procedure

#### Procedure: Wilcoxon Signed Rank Procedure

1. Calculate absolute differences:  $|X_i - \theta_0|$ .
2. Assign ranks  $R_i$  to these absolute differences (rank 1 = smallest).
3. Maximum sum of ranks:  $\sum R_i = \frac{n(n+1)}{2}$ .
4. Test Statistic:

$$T = \sum R_i Y_i$$

where  $Y_i$  is the indicator of positive sign (as defined for the Sign Test):

$$Y_i = \begin{cases} 1 & \text{if } X_i > \theta_0 \\ 0 & \text{if } X_i \leq \theta_0 \end{cases}$$

### 1.4.2 Worked Example: Heart Rate

#### Example: Wilcoxon Signed Rank Heart Rate Data

$$H_0 : \text{Median} = 70 \quad \text{vs} \quad H_a : \text{Median} > 70, \quad \theta_0 = 70.$$

**Data** ( $X_i$ ): 73, 82, 87, 68, 106, 60, 97.

| $X_i$ | $X_i - \theta_0$ | Sign ( $Y_i$ ) | $ X_i - \theta_0 $ | Rank ( $R_i$ ) | $R_i Y_i$ |
|-------|------------------|----------------|--------------------|----------------|-----------|
| 73    | +3               | 1              | 3                  | 2              | 2         |
| 82    | +12              | 1              | 12                 | 4              | 4         |
| 87    | +17              | 1              | 17                 | 5              | 5         |
| 68    | -2               | 0              | 2                  | 1              | 0         |
| 106   | +36              | 1              | 36                 | 7              | 7         |
| 60    | -10              | 0              | 10                 | 3              | 0         |
| 97    | +27              | 1              | 27                 | 6              | 6         |

**Test Statistic:**

$$T = \sum R_i Y_i = 2 + 4 + 5 + 7 + 6 = 24.$$

### 1.4.3 Distribution of $T$

- Let  $Y = \sum_{i=1}^n Y_i$  (number of positive signs).
- $T = \sum R_i Y_i$  is the sum of ranks corresponding to positive values of  $X_i - \theta_0$ .
- Let  $r_1 < \dots < r_y$  be the ranks corresponding to positive differences; vector of signs  $(r_1, \dots, r_n)$ .
- $Y \sim \text{Bin}(n, 0.5)$ .
- Under  $H_0$ , the vector of signs  $(Y_1, \dots, Y_n)$  can take any of the  $2^n$  values (from  $00 \dots 0$  to  $11 \dots 1$ ) with equal probability  $\frac{1}{2^n}$ .

### 1.4.4 Exact Null Distribution

Each configuration of signs gives a value for  $T$ . We count how many configurations sum to  $K$  to find  $P(T = K)$ .

#### Example: Exact Distribution, $n = 7$ : Finding $P(T = 9)$

Find all subsets of ranks  $\{1, 2, 3, 4, 5, 6, 7\}$  that sum to 9:

**Subsets of length 2:**  $(2, 7), (3, 6), (4, 5)$

**Subsets of length 3:**  $(1, 2, 6), (1, 3, 5), (2, 3, 4)$

Total combinations = 6.

$$P(T = 9) = \frac{6}{2^7}$$

#### Example: Other Probabilities for $n = 7$

$$P(T = 0) = \frac{1}{2^n} \quad (\text{empty set, all signs 0})$$

$$P(T = 1) = \frac{1}{2^n} \quad (\text{set } \{1\})$$

$$P(T = 2) = \frac{1}{2^n} \quad (\text{set } \{2\})$$

$$P(T = 3) = \frac{2}{2^n} \quad (\text{sets } \{3\} \text{ and } \{1, 2\})$$

### 1.4.5 Comparison of Tests

| Test             | Assumption                    | Use               |
|------------------|-------------------------------|-------------------|
| $t$ -test        | $F \equiv \mathcal{N}$        | One-sample centre |
| Sign test        | $F$ is continuous             | One-sample centre |
| Signed Rank test | $F$ is symmetric & continuous | One-sample centre |

All three tests maintain the **same significance level**  $\alpha$ .

#### Note

- If the assumption of **symmetry holds**, the Signed Rank test is more powerful than the Sign test.
- If the assumption of **symmetry does not hold**, the p-value obtained from the Signed Rank test will be *incorrect*.

## 1.5 Large-Sample Approximation for the Signed Rank Test

### Theorem: Distribution Parameters for $T$

$$\mathbb{E}[T] = \frac{n(n+1)}{4}$$

$$\text{Var}(T) = \frac{n(n+1)(2n+1)}{24}$$

**Standardised Statistic:**

$$T^* = \frac{T - \mathbb{E}[T]}{\sqrt{\text{Var}(T)}} \approx \mathcal{N}(0, 1)$$

### 1.5.1 Derivation of Mean and Variance

Let  $U_i = R_i Y_i$ . The ranks are fixed numbers  $1, \dots, n$  but  $Y_i$  is random. We view this as:

$$T = \sum_{i=1}^n U_i$$

Define  $V_i$  as the contribution of the  $i$ -th rank to the total sum  $T$ :

$$V_i = \begin{cases} i & \text{with probability } \frac{1}{2} \quad (\text{sign is positive}) \\ 0 & \text{with probability } \frac{1}{2} \quad (\text{sign is negative}) \end{cases}$$

**Abstract interpretation:** We define  $V_i$  as the contribution of the  $i$ -th rank to the total sum  $T$ . Instead of summing randomised ranks, we look at the fixed rank value  $i$  (e.g., rank 1, rank 2, rank 5) and ask: *Will this rank be added to our total?*

- If the sign is positive: we add the rank value  $i$  to our total (Value =  $i$ ).
- If the sign is negative: we do not add anything (Value = 0).

Since the sign is a “coin flip” (50/50 chance) under  $H_0$ :  $V_i$  takes the value  $i$  with probability  $1/2$ ;  $V_i$  takes the value 0 with probability  $1/2$ .

$$\mathbb{E}[V_i] = i \cdot \frac{1}{2} + 0 \cdot \frac{1}{2} = \frac{i}{2}$$

$$\mathbb{E}[T] = \sum_{i=1}^n \frac{i}{2} = \frac{1}{2} \cdot \frac{n(n+1)}{2} = \frac{n(n+1)}{4}$$

$$\text{Var}(V_i) = \mathbb{E}[V_i^2] - (\mathbb{E}[V_i])^2 = \frac{i^2}{2} - \frac{i^2}{4} = \frac{i^2}{4}$$

$$\text{Var}(T) = \sum_{i=1}^n \frac{i^2}{4} = \frac{1}{4} \sum_{i=1}^n i^2 = \frac{1}{4} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{n(n+1)(2n+1)}{24}$$

**Note**

- $V_i$  are independent because  $Y_i$  are independent.
- This approximation is useful when  $n$  is moderately large.
- Computation of  $2^n$  possibilities is very time-consuming for large  $n$ .

## 1.6 Handling Ties

### 1.6.1 Case 1: Zeros

- If  $X_i - \theta_0 = 0$  for some  $i$ .
- **Method:** Drop those observations and reduce  $n$ .

### 1.6.2 Case 2: Equal Absolute Differences

- If  $|X_i - \theta_0| = |X_j - \theta_0|$  for  $i \neq j$ .
- **Method:** Assign the **average rank**.

**Example: Average Rank Calculation**

Values to rank: 5, 6, 5, 6, 7.

1. Sort them: 5, 5, 6, 6, 7.

2. Initial positions: 1, 2, 3, 4, 5.

3. Average ties:

- The two 5s are at positions 1 and 2. Average rank =  $\frac{1+2}{2} = 1.5$ .
- The two 6s are at positions 3 and 4. Average rank =  $\frac{3+4}{2} = 3.5$ .
- The 7 is at position 5. Rank = 5.

4. Final Ranks: 1.5, 1.5, 3.5, 3.5, 5.

**Note**

When ties are present and average ranks are used, the p-value derived is **approximate**, not exact.

## 1.7 R Code: Week 1

```
# This generates all possible subsets of size 2 when n = 4.
combn(c(1:4), 2)

# Writing own function for p-value of signed rank test
signrank <- function(n, T){
  x <- c(1:n)
  k <- 0
  for(i in 1:n){
    y <- combn(x, i)
    l <- dim(y)[2]
    for(j in 1:l){
      k <- cbind(k, sum(x[y[,j]]))
    }
  }
  pval <- length(which(k >= T)) / length(k)
  return(pval)
}

table(k)

# Signed rank test for the example done in class
a <- c(73, 82, 87, 68, 106, 60, 97)
b <- a - 70
r <- rank(abs(b))
s <- ifelse(b > 0, 1, 0)
T <- sum(r * s)
signrank(7, T)

# Directly using built-in function for signed rank test
library(stats)
wilcox.test(a, mu = 70, alternative = "greater")
```

# Chapter 2

## One-Sample Estimation and Two-Sample Tests

### 2.1 One-Sample Estimation of Centre

Last week we learned about tests for the centre of one sample the non-parametric equivalent of the  $t$ -test. Now we move to **estimation**. The point estimate of the population median can be taken as the sample median. The question is about *interval* estimation. We also know from general inference that hypothesis tests can be inverted to obtain confidence intervals.

#### 2.1.1 Derivation of Interval Estimates

##### Model Setup

Let  $\theta$  be the population median.

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F, \quad \text{med}(F) = \theta.$$

##### Hypothesis and Test Statistic

Given the null hypothesis  $H_0 : \theta = a$ , the test statistic  $Y$  is defined as:

$$Y = \#\{X_i < a\} \sim \text{Bin}(n, \frac{1}{2}) \quad \text{under } H_0.$$

Do not reject  $H_0$  if:

$$\text{qbinom}(\frac{\alpha}{2}, n, 0.5) < Y < \text{qbinom}(1 - \frac{\alpha}{2}, n, 0.5).$$

##### Acceptance Region

The bounds for the acceptance region are determined by the quantiles of the Binomial distribution. The observed acceptance region is specified as:

$$179 < Y < 218.$$

##### Confidence Interval Construction

Using the count from the data:  $\#\{X_i < a\} = 179$ .

We invert the test to find the bounds for the parameter  $a$ :

$$a = X_{(179)}.$$

Thus the Confidence Interval (CI) is:

$$CI = (X_{(179)}, X_{(218)}).$$

### Theorem: CI from Inverting the Test

$(X_{(L)}, X_{(U)})$  is the CI for  $a$ , where  $(L < Y < U)$  is the acceptance region of the test.

## 2.1.2 Example: Academic Salary Data

### Example: Faculty Salary Confidence Interval

The 200809 nine-month academic salary for Assistant Professors, Associate Professors, and Professors in a U.S. college. Data collected as part of the ongoing effort of the college's administration to monitor salary differences between male and female faculty members.

```
library(carData)
data("Salaries")
qbinom(0.025, 397, 0.5)           # gives 179
qbinom(0.975, 397, 0.5)           # gives 218
sort(Salaries$salary)[c(179, 218)] # gives (105000,
111350)
```

What is the  $a$  so that  $k$  many  $X_i$  are less than  $a$ ? It is the  $k$ -th order statistic. So the confidence interval is  $(X_{(179)}, X_{(218)})$ .

When  $n$  is large, the `qbinom` quantities can be approximated by  $n/2 \pm \Phi^{-1}(\alpha/2)\sqrt{n/4}$ , since under  $H_0$ ,  $\mathbb{E}(Y) = n/2$  and  $\text{Var}(Y) = n/4$ .

## 2.2 Two-Sample Test for Centre

Now we start testing for equality of centre of two samples the non-parametric equivalent of the two-sample  $t$ -test when the parent distributions are not normal and sample sizes are small.

### 2.2.1 Setup

Let  $X_1, \dots, X_{n_1} \stackrel{\text{i.i.d.}}{\sim} D_1$  and  $Y_1, \dots, Y_{n_2} \stackrel{\text{i.i.d.}}{\sim} D_2$ .

**Hypothesis:**

$$H_0 : D_1 \equiv D_2 \quad (\text{Distributions are identical})$$

**Alternatives:**

$$H_a : D_1 \text{ is shifted to the right of } D_2 \quad (\text{right-tailed})$$

$$H_a : D_1 \text{ is shifted to the left of } D_2 \quad (\text{left-tailed})$$

$$H_a : D_1 \text{ is shifted right or left of } D_2 \quad (\text{two-sided})$$

**Assumptions:**

- Continuous distributions.
- Same shape (shift model).
- Independence of data values.
- Samples must be random and independent.

## 2.3 Wilcoxon Rank Sum Test

**Definition: Wilcoxon Rank Sum Test**

A non-parametric test for comparing the centres of two independent samples, using ranks of the pooled sample. Without loss of generality, let  $n_2 < n_1$ .

### 2.3.1 Procedure

**Procedure: Wilcoxon Rank Sum Algorithm**

1. Pool together  $X$ 's and  $Y$ 's.
2. Order the pooled sample and assign ranks.
3. Find the sum  $X$  of ranks of the  $X_i$ 's in the pooled sample.
4. The test statistic is  $T = X - \frac{n_1(n_1 + 1)}{2}$ .

**Logic:**

- If  $X$ 's are larger than  $Y$ 's (right-tailed alternative), then  $D_1$  is shifted to the right of  $D_2$ .
- Consequently, the *average ranks* of  $X$ 's will be larger than the average ranks of  $Y$ 's.

**Test Statistic:**

$$T = \text{sum of ranks of } X_i \text{'s in the pooled sample} - \frac{n_1(n_1 + 1)}{2}$$

$$T + T_1 = \frac{(n_1 + n_2)(n_1 + n_2 + 1)}{2} - \frac{n_1(n_1 + 1)}{2} = C$$

where  $T_1$  is the sum of ranks of  $Y_i$ 's. Therefore:

$$T_1 = C - T.$$

### 2.3.2 Numerical Example: Reaction Times

#### Example: Reaction Time Under Drugs A and B

We compare reaction times under the influence of two drugs (or treatments).

$$X = (2.07, 2.11, 2.43, 2.50, 2.71, 2.84, 2.88), \quad n_1 = 7$$

$$Y = (1.62, 1.71, 1.93, 1.96, 2.24, 2.41), \quad n_2 = 6$$

**Ranks after pooling and sorting:**

- Y's occupy ranks: 1, 2, 3, 4, 7, 8.
- X's occupy ranks: 5, 6, 9, 10, 11, 12, 13.

**Statistic Calculation:**

$$T = 5 + 6 + 9 + 10 + 11 + 12 + 13 = 66$$

$$U = 66 - \frac{n_1(n_1 + 1)}{2} = 66 - \frac{7 \times 8}{2} = 66 - 28 = 38.$$

**Result:** P-value = 0.01399 (two-tailed). At  $\alpha = 5\%$ , we **reject**  $H_0$ , i.e.,  $D_1 \neq D_2$ .  
With  $T = 38$ ,  $m = 7$ ,  $n = 13$  in R:

```
# The ranks of Y in the pooled sample are 5, 6, 9, 10,
# 11, 12, 13.
# So T = 38, m = 7, n = 13.
# The p-value obtained from R is 0.01399.
```

### 2.3.3 Null Distribution Construction

How do we find the distribution of the test statistic under  $H_0$  ( $D_1 \equiv D_2$ )?

#### Procedure: Permutation Method for Null Distribution

1. Generate all combinations of size  $n_1$  from the pooled set of size  $n_1 + n_2$ .
2. Total number of combinations:  $\binom{n_1 + n_2}{n_1}$ .
3. For each combination, label as "Group X" and calculate  $T$ .
4. Since  $H_0$  implies exchangeability, all combinations are equally likely (probability =  $1/\binom{n_1 + n_2}{n_1}$ ).
5. This generates the "Null Distribution".
6. Find the position of the observed  $T$  in this distribution to get the p-value.

## 2.4 Permutation Tests

### Definition: Permutation Tests

Tests that use permutations to obtain the null distribution. This technique can be applied to *any* test statistic even parametric statistics without requiring distributional assumptions.

### 2.4.1 Application to the $t$ -Statistic

Consider the two-sample  $t$ -statistic:

$$t = \frac{\bar{x} - \bar{y}}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

We do not know the distribution of this statistic unless  $X$ 's and  $Y$ 's are Normal. However, we can generate the distribution using the Permutation Technique.

### Procedure: Permutation $t$ -Test Procedure

1. Shuffle the data labels.
2. Re-calculate the  $t$ -statistic for every shuffle.
3. This empirical distribution serves as the null distribution, allowing the test without assuming Normality.

### Remark

There is no reason whatsoever why a permutation test has to use any particular test statistic. **Any test statistic will do!**

## 2.5 R Code: Week 2

```
# Wilcoxon rank sum test
# Example: Hollander & Wolfe (1973)
# Permeability constants of the human chorioamnion (a
# placental membrane) at
# term (x) and between 12 to 26 weeks gestational age (y).
# Alternative: greater permeability at term.
x <- c(0.80, 0.83, 1.89, 1.04, 1.45, 1.38, 1.91, 1.64, 0.73,
      1.46)
y <- c(1.15, 0.88, 0.90, 0.74, 1.21)
z <- sort(c(x, y))
T <- sum(which(z %in% x)) - 10*11/2
T

ranksum <- function(n, m, T){
  T <- T + m*(m+1)/2
  x <- c(1:n)
```

```
l <- choose(n, m)
k <- rep(0, l)
y <- combn(x, m)
for(j in 1:l){
  k[j] <- sum(x[y[,j]])
}
pval <- 2 * length(which(k < T)) / choose(n, m)
pval <- ifelse(pval > 1, 2 - pval, pval)
return(pval)
}

ranksum(15, 10, 35)

wilcox.test(
  c(0.80, 0.83, 1.89, 1.04, 1.45, 1.38, 1.91, 1.64, 0.73,
    1.46),
  c(1.15, 0.88, 0.90, 0.74, 1.21)
)

# Permutation test using t-statistic
T <- t.test(x, y)$statistic

t.pval <- function(a, b, T){
  c <- cbind(a, b)
  m <- length(b)
  n <- length(c)
  x <- c(1:n)
  l <- choose(n, m)
  t <- rep(0, l)
  y <- combn(x, m)
  for(j in 1:l){
    t[j] <- t.test(c[y[,j]], c[-y[,j]])$statistic
  }
  pval <- 2 * length(which(t >= T)) / l
  pval <- ifelse(pval > 1, 2 - pval, pval)
  return(pval)
}

t.pval(x, y, T)
```

# Chapter 3

## Mann-Whitney U-Statistic, General Score Functions, and Scale Tests

### 3.1 Wilcoxon Rank Sum and Mann-Whitney Equivalence

#### Definition: Wilcoxon Rank Sum Statistic

$$T = \sum_{i=1}^{n_1} R_{X_i} - \frac{n_1(n_1 + 1)}{2}$$

where  $R_{X_i}$  is the rank of  $X_i$  in the pooled sample  $\{X_1, \dots, X_{n_1}, Y_1, \dots, Y_{n_2}\}$ .

#### Definition: Mann-Whitney U-Statistic

$$T^* = \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} \mathbf{1}(X_i > Y_j)$$

#### Theorem: Equivalence of $T$ and $T^*$

$$T = T^*$$

Derivation:

$$T = \sum_{i=1}^{n_1} (R_{X_i} - i) = \sum_{i=1}^{n_1} \#\{y_j < x_{(i)}\}$$

Rank Definition Logic:

$$R_{X_{(i)}} = \#\{\text{samples} < x_{(i)}\} = i + \#\{y_j < x_{(i)}\}$$

Thus  $T = T^*$ , where  $T^*$  is the Mann-Whitney U-statistic.

#### 3.1.1 Properties of the U-Statistic

- The U-statistic is a **sum of unbiased estimators**.
- Let  $\theta = \mathbb{E}[\mathbf{1}(X_{(i)} > Y_{(j)})] = P(X > Y)$  for some parameter  $\theta$ .

- $\mathbb{E}[T^*] = n_1 n_2 \theta$ .

Under the Null Hypothesis ( $H_0$ ):

$$\theta = \frac{1}{2}$$

$$\mathbb{E}[\hat{\theta}] = \frac{n_1 n_2}{2}$$

$$\text{Var}(\hat{\theta}) = \frac{n_1 n_2 (n_1 + n_2 + 1)}{12}$$

### 3.1.2 Asymptotic Normality

A general procedure exists for finding the variance of U-statistics and establishing asymptotic normality:

$$Z = \frac{T^* - \frac{n_1 n_2}{2}}{\sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}} \xrightarrow{d} \mathcal{N}(0, 1)$$

(This is the two-sample test for centre using the large-sample approximation.)

## 3.2 General Linear Rank Statistics

### 3.2.1 General Form

#### Definition: General Linear Rank Statistic

The general form of rank statistics is:

$$\alpha \sum R_i + \beta.$$

More specifically, the **General Linear Rank Statistic**  $T_1$  is:

$$T_1 = \sum_{i=1}^{n_1} \varphi\left(\frac{R_i}{N+1}\right)$$

where  $N = n_1 + n_2$  is the total sample size and  $\varphi(u)$  is a **score function** defined on  $(0, 1)$ .

### 3.2.2 Assumptions on the Score Function $\varphi$

- $\int_0^1 \varphi(u) du = 0 \implies \mathbb{E}[T_1] = 0$
- $\int_0^1 \varphi^2(u) du = 1 \implies \text{Var}(T_1) \approx 1$  (scale normalisation)

For general scores, we define the process:

$$S_\varphi(\Delta) = \frac{1}{n} \sum_{j=1}^{n_2} a_\varphi[R(Y_j - \Delta)]$$

where  $R(Y_j - \Delta)$  denotes the rank of  $Y_j - \Delta$  among  $X_1, \dots, X_{n_1}$  and  $Y_1 - \Delta, \dots, Y_{n_2} - \Delta$ . A test statistic for  $H_0 : \Delta = 0$  vs  $H_A : \Delta > 0$  is  $S_\varphi(0)$ .

The generated scores are  $a_\varphi(i) = \varphi[i/(n+1)]$ , where  $n = n_1 + n_2$ . The Wilcoxon score function is  $\varphi_W(u) = \sqrt{12}[u - \frac{1}{2}]$ .

The mean and variance of  $S_\varphi(0)$  under the null hypothesis are:

$$\bar{a} = \frac{1}{n} \sum_{i=1}^n a_\varphi(i), \quad \frac{n_1 n_2}{n(n-1)} \sum_{i=1}^n (a_\varphi(i) - \bar{a})^2.$$

### 3.3 Measure of Scale

#### Remark

There is no unique measure of scale for a general distribution (unlike the median, which is the non-parametric alternative to the mean).

A general measure of scale for a distribution is:

$$(\mathbb{E}[|X - \mu|^k])^{1/k}, \quad M = \text{Median}.$$

### 3.4 Test for Comparing Scales of Two Populations

#### 3.4.1 Setup

- $X_1, \dots, X_{n_1} \sim F_X$
- $Y_1, \dots, Y_{n_2} \sim F_Y$

**Location-Scale Model:**

$$\text{pdf of } X = f_X\left(\frac{x - \mu_1}{\sigma_1}\right), \quad \text{pdf of } Y = f_Y\left(\frac{x - \mu_2}{\sigma_2}\right).$$

Same form  $f$ , different centre and scale. Let  $n_1 = 21$  (control) and  $n_2 = 22$  (experimental).

**Hypothesis:**

$$H_0 : \sigma_1 = \sigma_2 \quad H_a : \sigma_1 > \sigma_2.$$

### 3.4.2 Fligner-Killeen (FK) Test

#### Procedure: Fligner-Killeen Test Strategy

1. Estimate  $\mu_1, \mu_2$  using sample medians.
2. Calculate absolute deviations (aligned samples):

$$u_i = |x_i - \hat{\mu}_1|, \quad v_j = |y_j - \hat{\mu}_2|.$$

That is:  $X_i^* = X_i - \text{med}_X$  and  $Y_j^* = Y_j - \text{med}_Y$ .

3. If  $\sum u_i \geq \sum v_j$ , then  $u_i$  is shifted relative to  $v_j$ .
4. Perform **Wilcoxon Rank Sum Test** on  $\{u_i\}$  and  $\{v_j\}$ .

#### Remark

Any monotonic increasing function can be used for the test. General rank statistics can be applied here with a different  $\varphi$  from the linear one. Equivalent to taking squares, absolute values, or  $\log |\cdot|$ .

A more common test is the **Fligner-Killeen (FK)** procedure using the normal score function:

$$\varphi_{FK}(u) = \left( \Phi^{-1} \left( \frac{u + 1}{2} \right) \right)^2$$

where  $\Phi$  is the standard normal CDF. This leads to the test statistic:

$$S_{FK} = \sum_{j=1}^{n_2} \left( \Phi^{-1} \left( \frac{R|Y_j^*|}{2(n+1)} + \frac{1}{2} \right) \right)^2.$$

### 3.4.3 Numerical Result

#### Example: Fligner-Killeen Test: Ozone Effect on Rats

Draper and Smith (1966) present an experiment on the effect of ozone on the weight gain of rats. Two groups of rats were selected.

- **Control group** ( $n_1 = 21$ ): ozone-free environment for 7 days.
- **Experimental group** ( $n_2 = 22$ ): ozone environment for 7 days.

At the end of 7 days, the gain in weight of each rat was recorded.

$$R_i = \text{rank of } u_i = |x_i - \hat{\mu}_1| \text{ in the pooled sample}, \quad S_\varphi = \sum_{i=1}^{n_1} \varphi(R_i).$$

#### Results:

- Statistic  $Z = 2.095$
- P-value = 0.036 (one-tailed)
- **Conclusion:** Reject  $H_0$  (one-sided test required).

**Confidence Interval** for the ratio of scales  $\sigma_2/\sigma_1$ :

$$\text{CI} = (1.002, 5.63).$$

```
> data(sievers)
> x <- with(sievers, weight.gain[group == 'Control'])
> y <- with(sievers, weight.gain[group == 'Ozone'])
> fk.test(x, y)
statistic = 2.095976 , p-value = 0.03608434
95 percent confidence interval:
1.002458 5.636436
Estimate: 2.377034
```



# Chapter 4

## Multiple Samples, CDF Inference, and Goodness of Fit

### 4.1 Multiple Samples

#### 4.1.1 Parametric Case

We have  $m$  samples drawn from  $m$  populations:

$$\begin{aligned} X_{11}, \dots, X_{1n_1} &\stackrel{\text{i.i.d.}}{\sim} F_1(\mu_1) \\ X_{21}, \dots, X_{2n_2} &\stackrel{\text{i.i.d.}}{\sim} F_2(\mu_2) \\ &\vdots \\ X_{m1}, \dots, X_{mn_m} &\stackrel{\text{i.i.d.}}{\sim} F_m(\mu_m) \end{aligned}$$

*( $m$  samples from  $m$  populations)*

$$H_0 : \mu_1 = \dots = \mu_m.$$

#### Note

**ANOVA** requires the assumption that  $F_i$  are normal.

#### 4.1.2 Non-Parametric Case

For general  $F_i$  testing for equality of the sample **medians**.

#### Procedure: Kruskal-Wallis Procedure

1. Put all  $X_{ij}$ 's together and **rank** the combined sample.
2. For sample  $j$  ( $j = 1, \dots, m$ ), take the corresponding ranks (elements coming from the  $j$ -th sample).
3. Sum those ranks:  $R_j$ ,  $j = 1, \dots, m$ .

Under  $H_0$ :

$$\frac{R_j}{n_j} \longrightarrow \text{these should be close.}$$

There should be **no difference** (or very little) between the average ranks of different groups.

The **test statistic** is the variance of  $R_j/n_j$ :

$$\sum_j n_j \left( \frac{R_j}{n_j} - \bar{R} \right)^2$$

where  $n = \sum_{j=1}^m n_j$  and

$$\bar{R} = \text{avg rank for all samples together} = \frac{n(n+1)}{2n} = \frac{n+1}{2}.$$

## 4.2 Kruskal-Wallis Test

### Theorem: Kruskal-Wallis Statistic

$$\begin{aligned} H &= \frac{12}{n(n+1)} \left[ \sum_{j=1}^m \frac{R_j^2}{n_j} - \frac{(n+1)^2 n}{4} \right] \\ &= \frac{12}{n(n+1)} \sum_{j=1}^m \frac{R_j^2}{n_j} - 3(n+1). \end{aligned} \quad (4.1)$$

$H \approx \chi_{m-1}^2$  (approximately);  $n_j$  should be large for each  $j$ .

The exact distribution of the test statistic can be derived by **permutation**.

$$n = n_1 + n_2 + \cdots + n_m, \quad H_1, \dots, H_K.$$

**Number of permutations:**

$$\frac{n!}{n_1! n_2! \cdots n_m!} \quad (\text{many permutations})$$

**Example:**

$$\binom{14}{5} \binom{9}{7} = \frac{14!}{5! 7! 2!}$$

1. Calculate  $H$  for each permutation.
2. Find the p-value from the position of the observed  $H$  in the histogram.

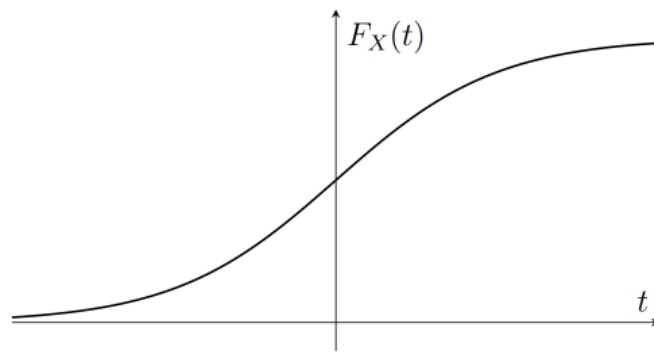


Figure 4.1

**Example: Kruskal-Wallis: Mucociliary Efficiency****Reference:** Hollander & Wolfe (1999).**Context:** Foam rate of diet in  $m = 3$  groups:

- Normal
- Obstructive airway
- Asbestosis

 $n_j : 5, 4, 5; \quad n = 14.$ **HW:** Write R code for this problem.

## 4.3 Inference for the CDF

**Definition: CDF as an Infinite-Dimensional Parameter**

The family  $\{F_x(b), -\infty < b < \infty\}$  is the parameter an **infinite-dimensional** parameter.

$X \rightarrow$  random variable;  $F_x(t) = P(X \leq t)$ ;  $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F_x$ .

For  $t = 5$ :  $P(X \leq 5) = \theta$  is estimated by  $\frac{1}{n} \#\{x_i \leq 5\}$ , giving estimate  $\hat{\theta}$ .

### 4.3.1 Empirical CDF

**Definition: Empirical CDF**

$$\hat{F}_n(t) = \frac{1}{n} \#\{x_i \leq t\}$$

For  $\theta = F_x(t)$ , a point estimate is  $\hat{\theta} = \hat{F}_n(t)$ .

**Distribution of  $\hat{\theta}$ :**

$$Y(t) = \#\{x_i \leq t\} = \sum_{i=1}^n \mathbf{1}(x_i \leq t) \sim \text{Bin}(n, \theta), \quad P(x_i \leq t) = F_x(t) = \theta.$$

If  $n$  is large enough, we can use the CLT for the Binomial.

**CI for  $\theta$ :**

$$\left( \hat{\theta} \pm z_{\alpha/2} \sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}} \right)$$

$$\hat{\theta} = \frac{Y(t)}{n} = \frac{1}{n} \# \{x_i \leq t\}.$$

#### Note

The estimate of  $F$  is a **step function** (discontinuous), even when  $F$  is continuous.

### 4.3.2 Order Statistics and the Empirical CDF

Given the order statistics  $X_{(1)} < \dots < X_{(n)}$ :

$$\hat{F}_n(t) = \begin{cases} 0 & t < X_{(1)} \\ \frac{1}{n} & X_{(1)} \leq t < X_{(2)} \\ \frac{2}{n} & X_{(2)} \leq t < X_{(3)} \\ \vdots & \\ \frac{n-1}{n} & X_{(n-1)} \leq t < X_{(n)} \\ 1 & X_{(n)} \leq t \end{cases}$$

Changes at  $n + 1$  points. No need to compute for every  $t$ .

CDF, PDF; quantile function  $F^{-1}(x)$ . Quantile  $\beta^{\text{th}}$ :  $\{F^{-1}(x)\}$ ,  $0 \leq x \leq 1$  a parameter.

### 4.3.3 Testing / Estimation of Median

When  $x = \frac{1}{2}$ :  $\theta = \text{median}$ ;  $\hat{\theta} = \text{sample median}$ .

$$H_0 : \theta = \theta_0 \quad \text{vs} \quad H_a : \theta \neq \theta_0.$$

$$\frac{\hat{\theta} - \theta_0}{\sqrt{\frac{\theta_0(1-\theta_0)}{n}}} \sim \mathcal{N}(0, 1).$$

Also derive an **exact test** for the median.

When  $x = \frac{3}{4}$ , the third percentile =  $\hat{\theta}$ :  $\hat{\theta} = \text{sample 3rd quartile}$ .

$$\hat{\theta} = \min\{x : 0.75n \text{ of the } x_i\text{'s are } \leq x\}.$$

CI and test for quantiles.

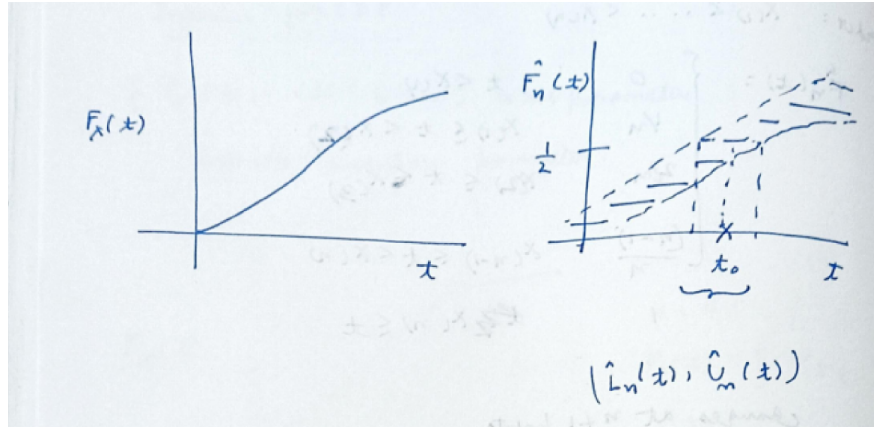


Figure 4.2: CDF and its Confidence interval

## 4.4 Confidence Intervals for CDF and Quantile Functions

### 4.5 Testing for Median (L-6)

#### 4.5.1 Is the Population Median 0.2?

Sample  $X_1, \dots, X_n$ .

$$H_0 : P(X > 0.2) = \frac{1}{2} = p_0 \quad H_a : p \neq p_0.$$

$$\hat{p} = \frac{1}{n} \# \{x_i > 0.2\}.$$

Under  $H_0$ :  $n\hat{p} \sim \text{Bin}(n, \frac{1}{2})$ .

Either **binomial test** or **normal approximation** if  $n$  is sufficiently large.

#### 4.5.2 Is the Third Quartile 0.3?

$$H_0 : P(X > 0.3) = 0.25 = p_0 \quad \text{or equivalently} \quad H_0 : P(X < 0.3) = 0.75.$$

$$H_a : p \neq p_0, \quad p = P(X > 0.3).$$

$$\hat{p} = \frac{1}{n} \# \{x_i > 0.3\}.$$

Under  $H_0$ :  $n\hat{p} \sim \text{Bin}(n, 0.25)$ .

P-value or critical region; use exact binomial or normal approximation.

### Example: Testing the Third Quartile

Given 176 observations out of 250 are less than  $a = 1.5$ . Test if 1.5 is the third quartile.

$$p = P(X < 1.5), \quad H_0 : p = 0.75, \quad H_a : p \neq 0.75, \quad \alpha = 5\%.$$

$$\hat{p} = \frac{176}{250}.$$

$$n\hat{p} \sim \text{Bin}(n, p_0) = \text{Bin}(250, 0.75).$$

$$Z = \frac{176 - 250 \times 0.75}{\sqrt{250 \times 0.75 \times 0.25}} = -1.68.$$

$$\text{p-value from } (-1.68) = 0.0929.$$

We do **not** reject at the 5%  $\alpha$ -level.

$$2 \times \text{pbinom}(176, 250, 0.75) = 0.1119 \quad (\text{exact}).$$

## 4.6 Estimating the CDF Further Details

Estimate CDF:  $F_x(t) = P(X \leq t) = \theta$ .

$$\hat{\theta} = \frac{1}{n} \# \{x_i \leq t\}.$$

$$n\hat{\theta} \sim \text{Bin}(n, \theta).$$

CI for  $\theta$ :

$$\left( \hat{\theta} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}} \right).$$

$$\hat{F}_n(t) = \begin{cases} 0 & t < X_{(1)} \\ 1/n & X_{(1)} \leq t < X_{(2)} \\ \vdots & \\ k/n & X_{(k)} \leq t < X_{(k+1)} \\ 1 & t > X_{(n)} \end{cases}$$

Not continuous; each step height is  $\frac{1}{n}$ .

## 4.7 Goodness of Fit: $\chi^2$ and KS Tests

Testing goodness of fit of a distribution.

$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F; H_0 : F = F_0.$

### 4.7.1 Case 1: $F_0$ Discrete $\chi^2$ Test

$$\chi^2 = \sum \frac{(\text{Obs} - \text{Exp})^2}{\text{Exp}}$$

**Binning:** histogram / frequency table. For continuous distributions, there is no obvious / objective choice of binning. Each bin has the same probability under  $H_0$ .

### 4.7.2 Kolmogorov-Smirnov Statistic

#### Definition: Kolmogorov-Smirnov (KS) Statistic

$$D = \max_{-\infty < t < \infty} |\hat{F}_n(t) - F_0(t)| = \max_{t=X_{(1)}, \dots, X_{(n)}} \{|\hat{F}_n(t) - F_0(t)|\}.$$

The maximum occurs at one of the data points; max step size =  $\frac{1}{n}$ .

#### Theorem: Properties of the KS Statistic

1. Evaluation is taken at **finitely many** points.
2.  $D$  is **distribution-free**.

### 4.7.3 Distribution-Free Property of $D$

$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F_0$ . Continuous distribution  $F_0$  increasing:  $F(X) \sim \text{Unif}(0, 1)$ .

Let  $S = \hat{F}_n(t) - F_0(t)$ :

$$\begin{aligned} S &= \frac{1}{n} \# \{x_i \leq t\} - F_0(t) \\ &= \#(i : F_0(x_i) \leq F_0(t)) \cdot \frac{1}{n} - F_0(t) \\ &= \#(i : U_i \leq F_0(t)) \cdot \frac{1}{n} - F_0(t), \quad U_1, \dots, U_n \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(0, 1) \\ &= \#(i : U_i \leq u) \cdot \frac{1}{n} - u, \quad [F_0(t) = u] \\ &= \tilde{F}_n(u) - u \end{aligned}$$

where  $\tilde{F}_n$  is the ECDF of  $U_1, \dots, U_n$ .

**Theorem: Distribution-Free Result**

The RHS does *not* depend on  $F_0$ , so  $D$  is **distribution-free**. It suffices to derive the distribution of  $D$  when  $F_0 = \text{Unif}(0, 1)$ .

# Chapter 5

## Density Estimation

### 5.1 Density Estimation

$f_x \rightarrow$  parameter (infinite dimensional parameter) of interest

1.  $f_x \geq 0$
2.  $\int f(x) dx = 1$
3.  $\int_a^b f(x) dx = P(a < X < b)$

$X_1, \dots, X_n \sim f_x$  have distribution same as  $X$ .

$\hat{f} \rightarrow$  histogram. Area under the curve as integral, then indeed  $\int \hat{f}(x) dx = 1$ .

$\hat{f} \geq 0$

For  $a, b$  endpoints of a bin,

$$\int_a^b \hat{f}(x) dx = \frac{\#\{X_i \in (a, b)\}}{n} = \hat{F}_n(b) - \hat{F}_n(a) = \text{empirical probability.}$$

Histogram satisfies (1)–(3).

#### Problems

1. It is not smooth.
2. Location and width of bins is subjective. (not a function of data only).

Minimize MSE to find optimal bin width (bandwidth).

Find MSE.  $x_0, x_1, x_2, \dots, x_n \rightarrow$  endpoints of bins.

$$\begin{aligned} \hat{f}(t) &= \begin{cases} \frac{\hat{F}_n(x + \delta) - \hat{F}_n(x)}{\delta} & x_i < t < x_{i+1} \\ 0 & \text{otherwise} \end{cases} \\ x_i &= x, \quad x_{i+1} = x + \delta \\ \hat{f}(t) &= \frac{\hat{F}_n(x + \delta) - \hat{F}_n(x)}{\delta} = \frac{\#\{X_i \in (x, x + \delta)\}}{n\delta} \end{aligned}$$

$$n\hat{F}_n(x) \sim \text{Bin}(n, F(x)) \quad \sim \frac{1}{n\delta} \text{Bin}(n, F(x + \delta) - F(x))$$

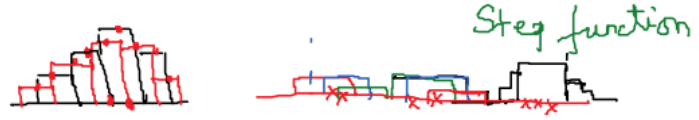


Figure 5.1: Step Function Diagram

$$F(x + \delta) - F(x) = p$$

$$E\hat{f}(t) = \frac{1}{n\delta} \cdot n[F(x + \delta) - F(x)] = \frac{1}{\delta}[F(x + \delta) - F(x)] = \frac{p}{\delta}$$

$$\text{Var } \hat{f}(t) = \frac{1}{n^2\delta^2} np(1-p) = \frac{p(1-p)}{n\delta^2}$$

$$\text{MSE} = \text{Bias}^2 + \text{Var} = \left(\frac{p}{\delta} - f(t)\right)^2 + \frac{p(1-p)}{n\delta^2}$$

Minimize thus over  $\delta$  to get bin width.

$$\left[ \begin{array}{l} \text{Best choice is } \delta \propto n^{-1/5}. \text{ Larger sample size should have} \\ \text{denser (more bins) histograms.} \\ \text{MSE} = O(n^{-4/5}) \text{ not } 1/n. \quad \text{MSE}(x) = \frac{\sigma^2}{n} \end{array} \right.$$

Location – Shifted histogram. (fix  $\delta$ ).

Actual evaluation does not need to be done at every ( $\mathbb{R}$ ) points, but only at discrete set of points.

$$\hat{f}(t) = \#\{X_i \in (t - \delta/2, t + \delta/2)\} / \delta$$

### Smoothness: Kernel density.

$K(z)$  is a standard kernel function if

- $K$  is Symmetric around zero  $k(z) = K(-z)$
- $K(z) \geq 0 \forall z \quad \int_{-\infty}^{\infty} K(z) dz = 1$
- The mass function is non decreasing on both sides of zero.
- $\int z k(z) dz = 0 \quad \int z^2 k(z) dz = 1$

Bandwidth re-scales kernel

$$k_h(z) = \frac{1}{h} K\left(\frac{z}{h}\right)$$

e.g. Rectangular  $K(z) = \frac{1}{2}\mathbf{1}(-1 < z < 1)$



Figure 5.2: smooth bump-shaped density estimate over data points  $x_1, x_2, \dots$

Gaussian  $K(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$

Triangular  $K(z) = (1 - |z|) \mathbf{1}(-1 < z < 1)$

Epanechnikov  $K(z) = \frac{3}{4}(1 - z^2) \mathbf{1}(-1 < z < 1)$

### Kernel density estimators

$$\hat{f}_h(t) = \frac{1}{n} \sum_{i=1}^n k_h(X_i - t)$$

Shifted histogram is a kernel density estimate with rectangular kernel. The height of the shifted histogram at a point  $t$  is proportional to the # of data points at a distance less than  $h$  to  $t$ .

Convolution of two densities  $f_x, g_y$

$$f_S(z) = f_x * g_y(z) = \int_{-\infty}^{\infty} f_x(t) g_y(z - t) dt = \int_{-\infty}^{\infty} f_x(z - t) g_y(t) dt$$

$f_S$  is the density of a r.v.  $S = X + Y$ .

Convolution of 2 pmfs  $p_x$  and  $p_y$

$$p_S(x) = p_x * p_y(x) = \sum_{t \in \mathbb{R}} p_x(t) p_y(x - t)$$

$p_S$  is the pmf of  $S = X + Y$

X independent of Y

$f_x$  is density of  $X$  and  $p_y$  is the pmf of  $Y$ .

$S = X + Y$  is continuous.

$$P(S \leq a) = \sum_{k \in \mathbb{R}} P(Y = k \text{ and } X \leq a - k) = \sum_{k \in \mathbb{R}} p_y(k) F_x(a - k)$$

$$f_S(a) = \sum_{t \in \mathbb{R}} p_y(t) f_x(a - t)$$

Back to kernels.

$$\hat{f}(t) = \frac{1}{n} \sum_{i=1}^n k_h(t - X_i) = \sum_{i=1}^n p_y(X_i) k_h(t - X_i)$$

$p_y$  is the pmf that puts mass  $\frac{1}{n}$  at each  $X_i, i = 1, \dots, n$ .

$$p_y(t) = \begin{cases} \frac{1}{n} & \text{if } t \in \{x_1, \dots, x_n\} \\ 0 & \text{o.w.} \end{cases}$$

So if  $k_h$  is density of a cont r.v. then  $\hat{f}(t)$  is also a density.

## Bandwidth selection

Minimize MSE.

$$\hat{f}_h(x) = \frac{1}{n} \sum_{i=1}^n k_h(x - X_i) \quad k_h(t) = \frac{1}{h} k\left(\frac{t}{h}\right) \quad \int_{\mathbb{R}} k(u) du = 1$$

$$\begin{aligned} E\left(\hat{f}_h(x)\right) &= \frac{1}{n} \sum_{i=1}^n E(k_h(x - X_i)) \quad X_i \sim f \\ &= E(k_h(x - X_1)) \end{aligned}$$

$$\text{Var}(\hat{f}_h(x)) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(k_h(x - X_i)) = \frac{1}{n} \text{Var}(k_h(x - X_1))$$

$$E(k_h(x - X_1)) = \int k_h(x - t) f(t) dt \quad k_h(x) \simeq 0 \text{ for } |x| > h$$

Taylor expansion  $f(t) = f(x) + (t - x) f'(x) + \frac{(t - x)^2}{2} f''(x) + o(h^2)$

$$\begin{aligned} E(k_h(x - X_1)) &= \int k_h(x - t) f(x) dt + \int k_h(x - t) (t - x) f'(x) dt + \int k_h(x - t) \frac{(t - x)^2}{2} f''(x) dt \\ &= f(x) \cdot 1 + f'(x) \int (t - x) k_h(t - x) dt + \text{smaller terms} \\ &\quad + \frac{f''(x)}{2} \int (t - x)^2 k_h(t - x) dt + \text{smaller terms} \end{aligned}$$

$$E(\hat{f}|x) = f(x) + 0 + \frac{h^2}{2} f''(x) + o(h^2)$$

$$\text{Bias} = \frac{h^2}{2} f''(x) + o(h^2).$$

$$\begin{aligned} E(k_h^2(x - X_1)) &= \int k_h^2(x - t) f(t) dt \\ &= \int \left[ \frac{1}{h} k\left(\frac{x - t}{h}\right) \right]^2 f(t) dt \\ &= \frac{1}{h^2} \int k^2\left(\frac{x - t}{h}\right) f(t) dt \quad \frac{x - t}{h} = u \\ &= \frac{1}{h^2} \cdot h \int k^2(u) f(x + hu) du \\ &= \frac{1}{h} \left[ \int k^2(u) f(x) du + \text{smaller terms} \right] \end{aligned}$$

$$\begin{aligned} \text{Var}(\hat{f}_h(x)) &= \frac{1}{n} \left[ \frac{1}{h} f(x) \cdot \int k^2(u) du + \text{smaller terms} - (E\hat{f})^2 \right] \\ &= \frac{1}{nh} \int f(x) \cdot k^2(u) du + \text{smaller terms.} \end{aligned}$$

$$\text{Bias} = \frac{h^2}{2} f''(x)$$

$$\text{MSE} = h^4 C_1(x) + \frac{1}{nh} C_2(x) + \text{smaller terms.}$$

$$h^4 = O\left(\frac{1}{nh}\right) \quad \text{To minimize MSE}$$

$$h \propto n^{-1/5}$$

$$h^4 = n^{-4/5}$$

$$\text{MSE} = O(n^{-4/5})$$

$$\frac{1}{nh} = n^{-1}h^{-1} = n^{-4/5}$$

$h < O(n^{-1/5})$  then  $\frac{1}{nh}$  is of higher order than  $n^{-4/5}$

$h > O(n^{-1/5})$  then  $h^4$  is of higher order than  $n^{-4/5}$

### Normal reference rule.

If  $f$  is normal then

$$h = 1.06 \sigma n^{-1/5}$$

= One can estimate  $\sigma$  from the data.

If the histogram is symmetric unimodal then this choice of  $h$  can be used.

### Cross-validation to determine $h$

$$E\left(\hat{f}_h(x) - f(x)\right)^2$$

minimise  $\int \left(\hat{f}_h(x) - f(x)\right)^2 dx$ .

$$\begin{aligned} &= \int \hat{f}_h^2(x) dx - 2 \int \hat{f}_h(x) f(x) dx + \int f^2(x) dx \\ &= F_1 + F_2 + F_3 \end{aligned}$$

For a particular  $h$ , we can compute  $\hat{f}_h(x)$  for every  $x$  and hence approximate the first term.

The third term does not contain  $h$ , so it does not affect the choice of  $h$ .

Second term involves  $h$  (so we do need it) and the unknown  $f$ .

$$-2 \int \hat{f}_h(x) f(x) dx = -2 E \hat{f}_h(X) \quad \text{where } X \sim f.$$

You have  $X_1, \dots, X_n$  i.i.d.  $f$ .

Estimate  $E \hat{f}_h(x)$  by  $\frac{1}{n} \sum_{i=1}^n \hat{f}_h(X_i)$ .

Then minimise the approximation to  $F_1$  (numerically) and (the estimate of  $F_2$ ) (numerically) to get  $h$ .

$$\boxed{\frac{1}{n} \sum_{i=1}^n \hat{f}_h(X_i)} \rightarrow \text{is biased for } E\hat{f}_h(x)$$

$$\frac{1}{n} \sum_{i=1}^n \frac{1}{n} \sum_{j=1}^n k_h(X_i - X_j)$$

$$E(k_h(X_i - X_j)) = E\hat{f}_h(X_i) \quad \text{for } i \neq j$$

$$i = j \quad E(k_h(X_i - X_j)) = E(k_h(0)) = k_h(0).$$

$$E \left[ \frac{1}{n} \sum_{j=1}^n k_h(X_i - X_j) \right] = \frac{1}{n} \left[ (n-1) E\hat{f}_h(x) + k_h(0) \right] = E\hat{f}_h(X_i) + \frac{1}{n} \left( k_h(0) - E\hat{f}_h(x) \right)$$

$$E \left[ \frac{1}{n} \sum_{j=1}^n k_h(X_i - X_j) \right] = \frac{1}{n} \left[ (n-1) E\hat{f}_h(x_i) + k_h(0) \right] = E\hat{f}_h(X_i) + \underbrace{\frac{1}{n} \left( k_h(0) - E\hat{f}_h(x) \right)}_{\text{bias.}}$$

Correct this by removing  $k_h(X_i - X_i)$

$$\hat{f}_{-i}(X_i) = \frac{1}{n-1} \sum_{j \neq i} k_h(X_i - X_j)$$

$$(-2) \frac{1}{n} \sum_{i=1}^n \hat{f}_{-i}(X_i) \rightarrow \text{estimates } F_2. \quad \text{Leave one out cross validation}$$

## 5.2 Bandwidth Selection

$f(x) \rightarrow$  parameter,  $y = f(x) + \varepsilon$

$\hat{f}(x) \rightarrow$  estimate,  $E[\varepsilon] = 0$

$$\int E \left[ f(x) - \hat{f}(x) \right]^2 dx \quad \text{MSE at } x.$$

$Y$  is integrated.

$$\text{MSE} = \text{Bias}^2 + \text{Var}$$

$$\text{bias} = O(h^2), \quad \text{var} = O\left(\frac{1}{nh}\right)$$

$$\text{MSE} = C_1 h^4 + C_2 \frac{1}{nh}$$

minimised at  $h \asymp n^{-\frac{1}{5}}$ .

**Optimal rate:**  $n^{-4/5}$ .

### 5.2.1 Cross-Validation

fix  $h$

#### Remark

**Leave one out.**

Given  $n$  points  $(x_1, y_1), \dots, (x_n, y_n)$ :

1. Take  $(x_i, y_i)$  out of the data.
2. Fit Kernel Regression on the rest of the points.
3. Compute

$$\sum_{i=1}^n \left( \hat{f}_{[-i]}(x_i) - y_i \right)^2 \longrightarrow \sum_{i=1}^n e_i^2.$$

This gives  $\widehat{\text{MSE}}(h)$ . Run this over values of  $h$  on a grid. Find  $h$  for which  $\widehat{\text{MSE}}(h)$  is minimised.

### 5.2.2 $K$ -Fold Cross-Validation for $h$

Given  $n$  points  $(x_i, y_i)$ ,  $i = 1, \dots, n$ .

#### Procedure: $K$ -Fold Cross-Validation Procedure

1. Divide  $n$  data points into  $K$  equal parts. Do this  $N$  times (e.g.  $N = 100$ ).
2. For each division, repeat for  $j = 1, \dots, K$ :
  - (a) Take the  $j$ -th part (chunk of size  $\lfloor n/K \rfloor$ ). Leave it out. Estimate  $f$  based on the remaining  $(K - 1)$  parts.
  - (b) Find the error on the  $j$ -th part by taking the average of squared errors:

$$\sum_{i \in \{j\}} \left( \hat{f}_{[-j]}(x_i) - y_i \right)^2.$$

3. Sum over  $j = 1, \dots, K$ .

This yields  $\widehat{\text{MSE}}(h)$ .

#### Note

- If  $h < O(n^{-1/5})$ , then  $\frac{1}{nh}$  is of higher order than  $n^{-4/5}$ .
- If  $h > O(n^{-1/5})$ , then  $h^4$  is of higher order than  $n^{-4/5}$ .

In either case, the MSE deteriorates; only  $h \asymp n^{-1/5}$  achieves the optimal rate  $O(n^{-4/5})$ .

#### Procedure: $K$ -Fold Cross-Validation

For  $h$ : for one division of  $n$  units into  $K$  equal parts,

1. **Repeat** for  $j = 1, \dots, K$ :
  - Take  $j^{\text{th}}$  part. Leave it out. Estimate  $f$  based on the  $(K - 1)$  parts.
  - Find the error on the  $j^{\text{th}}$  part, by taking average of squared errors.
2. Sum to determine  $h$ .

Repeat  $N$  times  $\rightarrow \widehat{\text{MSE}}(h)$ .